

Case interview transcript — Brew & Bean Profitability

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Interviewer: Cerebras gpt-oss-120b, reasoning high (experiment) · Candidate: default simulated candidate, human pacing · not scored

Interviewer · 0:00 (turn 0)

Your client is Brew & Bean, a specialty coffee chain with 200 locations across the US. Over the past two years, their net profit margin has declined from 24% to 6%, even though revenues have grown 15% over the same period. The CEO has hired us to identify the root cause of this profit decline and recommend how to reverse it. How would you approach this problem?

Take a minute if you need it, then tell me how you'd structure your approach.

Candidate · 1:30 (turn 1)

Let me take a second to lay this out.

So revenue is up 15% but margin fell from 24% to 6% — that's a huge drop, and since revenue grew, the problem is almost certainly on the cost side, or a mix issue where we're growing low-margin revenue. I'd want to confirm that first.

My structure has three buckets.

First, revenue quality: is the 15% growth coming from more stores, more transactions per store, or higher prices? If it's new store openings, those stores may be dragging margin. I'd also want to know if ticket size changed, and whether we discounted or shifted mix toward lower-margin items.

Second, costs — I'd split into COGS (coffee beans, milk, cups) and operating costs (labor, rent, utilities, marketing, corporate overhead). I want the cost breakdown as a percent of revenue for both years so I can see which line moved.

Third, external factors: commodity prices, wage legislation, competition.

Can I start with the cost breakdown by year?

Interviewer · 1:31 (turn 2)

Okay. I'll come back to cost breakdown by year shortly. Is that framework MECE — what's missing?

Candidate · 3:01 (turn 3)

Fair push. Let me check it.

A couple of gaps. First, I jumped to cost without being precise about profit: profit is revenue minus cost, so technically my top level should be those two branches, and the revenue branch itself splits into price times volume, with volume splitting into number of stores times transactions per store times ticket. I sort of mixed those levels together.

Second, I put "external factors" as a third bucket, but that's not really parallel — commodity prices and wages are drivers of cost lines, not a separate branch. It's a cause, not a category. So that's overlapping.

Third, I didn't account for anything below the operating line. Net margin, not gross — so interest expense, depreciation from new store buildouts, taxes, or one-time write-offs could all be hitting us. If they opened a lot of stores, depreciation and debt service could be a real chunk.

So cleaner: profit equals revenue minus operating costs minus non-operating costs, with price and volume under revenue.

Interviewer · 3:07 (turn 4)

Understood. Is that MECE — what's missing?

Candidate · 4:37 (turn 5)

I think the bigger gap now is that I've only structured it by line item, not by segment. The same framework needs a cutting dimension — otherwise averages hide everything.

So I'd layer in: new stores versus mature stores, since new ones likely open at low volume with full fixed costs; geography, since wages and rent vary a lot by metro; and channel or product mix — in-store versus delivery apps, which take big commissions, and food versus beverage, which have very different margins.

I'd also add a time dimension. A two-year decline could be gradual or a cliff. If margin fell off suddenly, that points to a discrete event like a contract renegotiation or a delivery partnership. If it eroded steadily, it's more structural.

Last thing I missed: capacity utilization. Revenue up 15% across 200 stores might mean each store is doing less volume against fixed rent and labor.

Can I get the cost breakdown by year now?

Interviewer · 4:38 (turn 6)

Got it. You asked about this earlier: COGS is 58% of revenue today, up from 42% two years ago. Labor is 22% of revenue, and it's been stable. Overhead is 14% of revenue, up slightly from 12%. What do these cost percentages reveal about the drivers of the margin decline?

Candidate · 6:08 (turn 7)

This basically solves the arithmetic. Let me check it adds up.

Today: 100 minus 58 COGS minus 22 labor minus 14 overhead = 6% margin. Two years ago: 100 minus 42 minus 22 minus 12 = 24%. So the full 18-point decline is explained by these lines — nothing hidden below the operating line after all.

And the split is lopsided: COGS is 16 of the 18 points, overhead only 2. So roughly 89% of the problem is COGS. Labor flat at 22% is actually notable — if wages rose industry-wide and labor held flat as a percent of revenue, they may have managed that well, or revenue growth absorbed it.

So I'd focus almost entirely on COGS. Two possibilities: input cost per unit went up — bean or dairy prices, supplier contracts — or price didn't keep pace, meaning they absorbed cost increases instead of passing them through. Mix shift toward lower-margin items could do it too.

Do we know whether menu prices changed over these two years, and what's inside that COGS line?

Interviewer · 6:09 (turn 8)

Got it. Menu prices have not changed in two years. Raw coffee bean costs are up 40% over the past two years. Coffee beans were 25% of COGS two years ago. I'll come back to other input cost changes (dairy, packaging, food) over 2 years shortly. Based on the menu price changes and COGS breakdown, which factor do you think is driving most of the COGS increase?

Candidate · 7:39 (turn 9)

Let me size the bean effect.

Beans were 25% of COGS, and COGS was 42% of revenue, so beans were $0.25 \times 42 = 10.5\%$ of revenue two years ago. Prices unchanged, so revenue per unit is flat — a 40% cost increase takes beans to $10.5 \times 1.4 = 14.7\%$ of revenue. That's a 4.2-point increase.

But total COGS rose 16 points. So beans explain only 4.2 of 16, about 26% — a quarter of the COGS problem. That surprises me; I'd assumed beans were the story.

So roughly 12 points of COGS increase are coming from somewhere else — dairy, packaging, food, or waste and spoilage. That's the bigger fish, and I'd want that breakdown next.

The flat menu pricing is the other half of the answer. With zero price increases over two years while inputs inflated, every cost increase drops straight to the bottom line. Even a modest pass-through would have protected a lot of margin.

What happened to dairy, packaging, and food costs?

Interviewer · 7:41 (turn 10)

Understood. On what you asked about earlier: Other input costs — dairy, packaging, and food — are up 37.5% over the past two years. Here's an exhibit: Brew & Bean Cost Structure Over Time. What does this breakdown tell you about the drivers of the COGS increase?

Candidate · 9:11 (turn 11)

The numbers close cleanly. Other inputs were 75% of COGS = $0.75 \times 42 = 31.5\%$ of revenue. Up 37.5%: $31.5 \times 1.375 = 43.3\%$, so +11.8 points. Add the 4.2 from beans and I get 16.0 points — exactly the COGS increase. So input cost inflation explains 100% of it. No waste, shrinkage, or mix story needed.

And the split is 4.2 beans versus 11.8 other, so about 74% of the COGS problem is dairy, packaging, and food — not coffee. That matters for the recommendation; renegotiating bean contracts alone touches only a quarter of the issue.

The exhibit tells me it's a steady erosion, not a cliff: COGS up 8 points, then 8 points again. Profit 24, 15, 6. That's structural inflation, not a one-time event — and it suggests a third straight year would push them near zero or negative.

The damning piece is still flat pricing through two years of roughly 38–40% input inflation. Do we know anything about competitor pricing or customer price sensitivity?

Interviewer · 9:13 (turn 12)

Got it. I don't have competitor pricing data or customer price sensitivity data. Go on.

Candidate · 10:43 (turn 13)

Let me size what pricing alone would have to do.

Total costs are $58 + 22 + 14 = 94\%$ of revenue. If I raise prices $x\%$ and hold volume constant, costs stay flat in absolute dollars while revenue scales. To get back to 24% margin: $94 \div 0.76 = 123.7$, so about a 24% price increase. That's steep and almost certainly costs volume.

A more realistic target — say 15%, their level last year — needs $94 \div 0.85 = 110.6$, roughly an 11% increase. That's broadly in line with input inflation of 38–40% on a cost base that's about 60% of revenue, so it's a pass-through, not a grab.

Worth noting: because labor and overhead are largely fixed, a price increase that holds volume improves margin faster than a cost cut of the same size.

Without elasticity data I'd assume specialty coffee is moderately inelastic given the premium positioning. But I'd want to test it rather than roll out chain-wide.

What else can we look at — anything on the supply side, contracts or sourcing?

Interviewer · 10:45 (turn 14)

Got it. I don't have supply contracts and sourcing details. Beyond pricing, what other actions could Brew & Bean pursue to offset the cost inflation and improve margins?

Candidate · 12:15 (turn 15)

Three other levers beyond price.

First, procurement. Even without contract details, 200 locations is real scale — I'd look at consolidating suppliers, committing to volume for discounts, and hedging or forward-contracting beans and dairy to stop the bleeding from spot prices. Dairy especially, since other inputs are three-quarters of the problem.

Second, product and mix. Reformulate where the customer won't notice — lighter cup stock, cheaper lids, portion control on milk. Push mix toward high-margin, low-input items: black coffee, espresso, and especially non-dairy or non-food attachments. If dairy is the inflation driver, steering customers toward drinks that use less of it helps twice.

Third, indirect pricing. Shrink drink sizes slightly at the same price, trim discounts and loyalty giveaways, or charge for add-ons that were free. Customers feel these less than a headline price increase.

I'd also sanity-check overhead, which drifted 12 to 14. It's only 2 points, but it's the one line moving in the wrong direction that isn't inflation-driven.

Interviewer · 12:16 (turn 16)

Got it. Beyond the levers you've mentioned, what other actions could Brew & Bean pursue to improve margins?

Candidate · 13:46 (turn 17)

A few more.

On the revenue side rather than cost: grow volume into the fixed base. Labor and overhead are 36% of revenue and largely fixed per store, so incremental transactions are highly accretive. Extend dayparts — afternoon and evening traffic, food attachment at lunch — and push catering or office accounts. Also consider new revenue

streams with better margins: selling retail bags of beans, merchandise, or subscriptions, which carry input costs but far less labor.

On labor, it's been flat at 22% so it's not a problem, but it's also an opportunity nobody's touched. Mobile ordering, better scheduling against traffic patterns, or some automation could pull a point or two out.

On footprint, I'd want store-level P&Ls. With 200 locations there's likely a tail of underperformers; closing or renegotiating leases on the worst decile could help. Rent sits in overhead, the line that's drifting.

Finally, delivery apps — if they're on them, those commissions are margin-destructive and worth revisiting.

Interviewer · 13:47 (turn 18)

Okay. Based on what you have, what would you tell the CEO to do?

Candidate · 15:17 (turn 19)

My recommendation: raise prices roughly 10–12% to restore margin to the mid-teens, and pair it with procurement action on dairy and packaging.

The reasoning. The entire 18-point margin decline sits in COGS — 16 of 18 points — and input inflation of 38–40% explains all 16. Labor held flat and overhead moved only 2 points. Meanwhile menu prices haven't moved in two years, so they've absorbed every dollar of inflation. A roughly 11% increase gets them to $94 \div 0.85$, about 15% margin, and it's a pass-through rather than a grab.

The second priority is dairy, packaging and food, which is 11.8 of the 16 points — three-quarters of the problem. Beans get the attention but aren't the story. Use 200-store scale to consolidate suppliers and forward-contract.

Risks: no elasticity or competitor data, so a chain-wide increase could cost volume. The trajectory is also 8 points of COGS per year, so inaction means near-zero profit next year.

Next steps: pilot pricing in 20 stores across two or three metros for six weeks, measure volume response, and run a dairy and packaging RFP in parallel.

Interviewer · 15:18 (turn 20)

Got it. What is the biggest risk to your recommendation?

Candidate · 16:48 (turn 21)

The biggest risk is price elasticity — I have zero data on it, and the whole recommendation rests on volume holding.

Let me size the cushion. Treat COGS as variable and labor plus overhead as fixed. Index revenue at 100, so variable cost is 58 and fixed is 36. With an 11% price increase, each unit of volume now earns 111 in revenue against 58 in variable cost, so contribution is 53. Profit equals $53 \times (\text{remaining volume share})$ minus 36.

To stay at today's 6%: $53 \times (1-v) = 42$, so $1-v = 0.79$, $v \sim 21\%$.

So they could lose about a fifth of their transactions and still be no worse off than today. That's a wide margin of safety, which makes me more comfortable.

The secondary risk is that this is a pure pass-through with no competitive insight — if rivals held prices, we're suddenly the expensive option and brand perception takes a hit. That's exactly what the 20-store pilot is designed to catch before a chain-wide rollout.

Interviewer · 16:49 (turn 22)

That's our time. Thanks for working through this with me — your written report will follow.